

Dittert's conjecture in dimension 9

An exact computer-assisted boundary-exclusion proof

Pedro Paulo Marques do Nascimento

23 July 2026

Status. This is an exact, reproducible working proof. It has not yet undergone independent peer review. The finite dimension-specific calculations are checked by two separately implemented standard-library verifiers.

Licensed under [CC BY 4.0](#).

Abstract

For a nonnegative $n \times n$ matrix A whose entries sum to n , let r_i, c_j be its row and column sums and put

$$\Phi(A) = \prod_i r_i + \prod_j c_j - \text{per } A.$$

We prove Dittert's conjecture in dimension $n = 9$. A maximal Li scaling reduces the boundary problem to whole, marginal, and proper tight cuts. The proper cuts are excluded by exact staircase-face bounds. The direct scalar estimate for a marginal cut has a small failure interval. On that interval, stationarity forces a large column containing the minimum row; cofactor nonnegativity bounds every other row meeting this column. Seven exact degree-82 Bernstein certificates then exclude every possible column support. The proof uses only a two-independent-zero permanent bound and does not assume a three-zero permanent-minimum theorem.

1 Statement and published inputs

Let

$$K_n = \left\{ A = (a_{ij}) \in \mathbb{R}_{\geq 0}^{n \times n} : \sum_{i,j} a_{ij} = n \right\}, \quad \gamma_n = \frac{n!}{n^n}.$$

For $A \in K_n$, write

$$R = \prod_{i=1}^n r_i, \quad C = \prod_{j=1}^n c_j, \quad P = \text{per } A.$$

Thus $\Phi(A) = R + C - P$ and $\Phi(J_n/n) = 2 - \gamma_n$.

Theorem 1.1. *For every $A \in K_9$,*

$$\Phi(A) \leq 2 - \frac{9!}{9^9},$$

with equality if and only if $A = J_9/9$.

We use the following published results.

1. Hwang proved that the only possible positive Φ -maximizer in K_n is J_n/n [5].
2. Cheon and Wanless proved that a partly decomposable matrix cannot maximize Φ , and that a maximizer cannot have its complete zero set equal, up to row and column permutations, to one proper rectangular block [1, Theorems 3.2 and 3.7].
3. Li's cut characterization says that a nonnegative $n \times n$ matrix S is doubly superstochastic if and only if

$$\sum_{i \in I, j \in J} s_{ij} \geq |I| + |J| - n$$

for all $I, J \subseteq [n]$ [6]; see also [1, Lemma 2.2].

4. On a staircase face of the Birkhoff polytope, the barycenter minimizes the permanent [4].
5. Minc's averaging theorem for permanent minimizers with repeated support rows and columns gives the block form used below [7, Theorem 1]; see also [8, Section 2].
6. The van der Waerden permanent theorem gives $\text{per } B \geq \gamma_n$ for every doubly stochastic B , with equality only at J_n/n [2, 3].

The support-stationarity equation and the large-column support argument used below are proved directly; they are not additional external assumptions.

2 Boundary structure and the shared deficit

Because K_n is compact, Φ has a global maximizer. By Hwang's theorem, a nonuniform maximizer must lie on the boundary.

Lemma 2.1 (Two independent zeroes). *Every boundary Φ -maximizer has two zero entries in distinct rows and distinct columns.*

Proof. Suppose a nonempty zero set contains no such pair. Fix a zero at (i, j) . Every other zero shares row i or column j . If there were both a zero (i, j') , $j' \neq j$, and a zero (i', j) , $i' \neq i$, those two zeroes would be independent. Hence all zeroes lie in one row or all lie in one column.

If that row or column is entirely zero, the matrix is partly decomposable. Otherwise its complete zero set is one proper rectangular block. Both cases are excluded for a maximizer by Cheon–Wanless. $\square$

Assume for contradiction that a boundary maximizer A satisfies $\Phi(A) \geq 2 - \gamma_n$. Since the row sums and column sums each have arithmetic mean 1, AM–GM gives $R, C \leq 1$. Consequently $P \leq \gamma_n$. Write

$$P = \gamma_n - \delta, \quad 0 \leq \delta \leq \gamma_n,$$

and put $R = 1 - \rho$, $C = 1 - \sigma$. The assumed lower bound gives

$$\rho + \sigma \leq \delta. \tag{1}$$

In particular,

$$R \geq 1 - \delta, \quad C \geq 1 - \delta, \quad RC \geq 1 - \delta. \tag{2}$$

3 The maximal Li scaling

For $I, J \subseteq [n]$, let $s(A[I, J])$ be the sum of the indicated submatrix, and define

$$\lambda = \min_{\substack{I, J \subseteq [n] \\ k := |I| + |J| - n > 0}} \frac{s(A[I, J])}{k}. \quad (3)$$

A maximizer is fully indecomposable by the Cheon–Wanless exclusion. Hence every numerator in (3) is positive, so $\lambda > 0$; taking $I = J = [n]$ shows $\lambda \leq 1$.

Li's theorem says that A/λ is doubly superstochastic. Thus there is a doubly stochastic matrix B such that

$$\lambda B \leq A \quad \text{entrywise}, \quad P \geq \lambda^n \text{ per } B. \quad (4)$$

Choose I, J attaining (3), and set

$$u = n - |I|, \quad v = n - |J|, \quad k = n - u - v > 0.$$

There are three cases: the whole cut $(u, v) = (0, 0)$, a marginal cut where exactly one of u, v is zero, and a proper cut where $u, v \geq 1$.

If $u = v = 0$, then $\lambda = 1$. Since $B \leq A$ and both matrices have total sum n , $A = B$ is doubly stochastic. The van der Waerden theorem and $P \leq \gamma_n$ force $A = J_n/n$, contrary to the boundary assumption.

4 The two-independent-zero permanent gap

Let $\Omega_n^{(2)}$ be the face of the Birkhoff polytope specified by $b_{12} = b_{21} = 0$, and put $m = n - 2$. Minc's averaging theorem permits a permanent minimizer on this face to be chosen in the form

$$Y = \begin{pmatrix} x_1 & 0 & a_1 \mathbf{1}^\top \\ 0 & x_2 & a_2 \mathbf{1}^\top \\ a_1 \mathbf{1} & a_2 \mathbf{1} & x J_m \end{pmatrix}, \quad x_i = 1 - m a_i, \quad a_1 + a_2 + m x = 1. \quad (5)$$

Here $0 \leq a_i \leq 1/m$.

Proposition 4.1 (One-variable reduction). *For $m \geq 4$, the minimum permanent in (5) occurs with $a_1 = a_2$. Consequently the minimum of per B on $\Omega_n^{(2)}$ is the minimum, on*

$$\frac{m-2}{m^2} \leq x \leq \frac{1}{m},$$

of

$$P_n(x) = m! x^{m-2} \left(x^2 a^2 + 2m x a b^2 + m(m-1) b^4 \right), \quad (6)$$

where

$$a = \frac{2-m+m^2 x}{2}, \quad b = \frac{1-mx}{2}. \quad (7)$$

Proof. Put $s = a_1 + a_2$, $p = a_1 a_2$, and $x = (1-s)/m$. Counting permanent terms in (5) gives, apart from the positive factor $m! x^{m-2}$,

$$E_m(s, p) = x^2(1 - ms + m^2 p) + mx(s^2 - (2 + ms)p) + m(m-1)p^2.$$

For fixed s ,

$$\frac{\partial E_m}{\partial p} = 2m(m-1)p + (m+1)s^2 - ms - 1.$$

Since $p \leq s^2/4$ and $0 \leq s \leq 2/m$, this is at most

$$D_m(s) = \frac{(m^2 + m + 2)s^2 - 2ms - 2}{2}.$$

The quadratic is convex, while

$$D_m(0) = -1, \quad D_m(2/m) = -1 + \frac{2}{m} + \frac{4}{m^2} < 0 \quad (m \geq 4).$$

Thus the derivative is strictly negative. For fixed s , the permanent is minimized by maximizing p , namely by $a_1 = a_2 = b$. Solving the stochastic constraints yields (7) and the stated interval. A second count gives (6). $\square$

For $n = 9$, set

$$L_9 = \frac{47533}{50000000}. \quad (8)$$

Proposition 4.2 (Exact two-zero bound). *Every $B \in \Omega_9^{(2)}$ satisfies*

$$\text{per } B > L_9 > \gamma_9.$$

Proof. Here $m = 7$ and the complete interval is $[5/49, 1/7]$. The primary verifier expands $P_9(x) - L_9$ over $\mathbb{Q}$, verifies positive endpoint values, and uses an exact rational Sturm sequence to prove that it has no root on the interval.

The independent verifier instead evaluates the matrix (5) with (7) by the literal exact Ryser formula at ten rational points, interpolates the degree-nine polynomial, and proves positivity by exact Bernstein coefficients on 256 subintervals. Finally,

$$L_9 - \gamma_9 = \frac{3348865477}{239148450000000} > 0. \quad (9)$$

$\square$

5 Marginal tight cuts

We first record the stationarity equation used twice below.

Lemma 5.1 (Support stationarity). *At a global maximizer, for every positive entry a_{ij} ,*

$$\frac{R}{r_i} + \frac{C}{c_j} - \text{per } A(i \mid j) = \Phi(A), \quad (10)$$

where $A(i \mid j)$ is obtained by deleting row i and column j .

Proof. The left side is $\partial\Phi/\partial a_{ij}$. Moving a small amount of mass between positive entries shows that all partial derivatives on the support have a common value μ . Euler's identity, using homogeneity of degree n , gives

$$n\Phi(A) = \sum_{i,j} a_{ij} \frac{\partial\Phi}{\partial a_{ij}} = \mu \sum_{i,j} a_{ij} = n\mu.$$

Thus $\mu = \Phi(A)$. $\square$

Suppose the tight cut is marginal. Its ratio in (3) is an average of a nonempty set of row sums or column sums. By transposition, it is enough to treat a minimum row. Choose i with

$$r_i = \lambda = 1 - a, \quad 0 \leq a < 1.$$

Multiply (10) by a_{ij} and sum over j . Expansion of the permanent along row i gives

$$R + C \sum_j \frac{a_{ij}}{c_j} - P = r_i \Phi(A).$$

Therefore

$$q_i := \sum_j \frac{a_{ij}}{c_j} = 1 - (1 + \theta)a, \quad \theta = \frac{R - P}{C}. \quad (11)$$

By (2) and $P = \gamma_n - \delta$,

$$\theta \geq R - P \geq 1 - \gamma_n. \quad (12)$$

Let $c_{\max} = \max_j c_j$. Since $q_i > 0$, equations (11)–(12) imply

$$\frac{1 - a}{c_{\max}} \leq q_i \leq 1 - (2 - \gamma_n)a.$$

Thus the last denominator is positive and

$$c_{\max} \geq 1 + h(a), \quad h(a) = \frac{(1 - \gamma_n)a}{1 - (2 - \gamma_n)a}. \quad (13)$$

AM–GM on the other eight row sums and on the other eight column sums gives

$$R \leq f(a) := (1 - a) \left(1 + \frac{a}{8}\right)^8, \quad (14)$$

$$C \leq g(h(a)) := (1 + h(a)) \left(1 - \frac{h(a)}{8}\right)^8. \quad (15)$$

Define

$$D(a) = 2 - f(a) - g(h(a)). \quad (16)$$

The shared deficit implies $\delta \geq D(a)$. Moreover, f decreases for $a > 0$, h increases, and g decreases for positive arguments, so D is strictly increasing. The exact check

$$D(3/100) > \gamma_9 \quad (17)$$

therefore confines every feasible value to $0 \leq a < 3/100$.

The dominated doubly stochastic matrix B inherits the two independent zeroes of A , so Proposition 4.2 gives

$$P \geq L_9(1 - a)^9. \quad (18)$$

The elementary scalar contradiction is

$$H(a) := L_9(1 - a)^9 - \gamma_9 + D(a) > 0. \quad (19)$$

After clearing the positive denominator $8^8[1 - (2 - \gamma_9)a]^9$, its numerator has degree 18. The primary verifier proves by exact Sturm sequences that $H > 0$ on

$$[0, 1/500] \cup [1/200, 3/100]. \quad (20)$$

The independent verifier reconstructs this numerator from exact values and proves the same two claims by Bernstein positivity. In view of (9), only the interval

$$\frac{1}{500} \leq a \leq \frac{1}{200} \quad (21)$$

requires a stronger argument.

5.1 The large-column support bound

The weighted-average identity

$$\frac{q_i}{r_i} = \sum_{j: a_{ij} > 0} \frac{a_{ij}}{r_i} \frac{1}{c_j}$$

shows that some j_* with $a_{ij_*} > 0$ satisfies

$$c_{j_*} \geq \frac{r_i}{q_i} \geq 1 + h(a). \quad (22)$$

The important extra fact, beyond (13), is that this large column contains the minimum row.

Put

$$d(a) = \left(1 - \frac{h(a)}{8}\right)^8, \quad q(a) = 2 - \gamma_9 - d(a). \quad (23)$$

For every ℓ with $a_{\ell j_*} > 0$, cofactor nonnegativity and Lemma 5.1 give

$$\frac{R}{r_\ell} = \Phi(A) - \frac{C}{c_{j_*}} + \text{per } A(\ell \mid j_*) \geq 2 - \gamma_9 - d(a) = q(a).$$

Indeed, C/c_{j_*} is the product of the other eight column sums, whose sum is at most $8 - h(a)$, so AM–GM bounds it by $d(a)$. On (21), q is increasing and the verifier checks exactly that $q(1/500) > 1$. Since $R \leq 1$, every row meeting j_* satisfies

$$r_\ell \leq \frac{R}{q(a)} \leq \frac{1}{q(a)} < 1. \quad (24)$$

Let z be the number of zero entries in column j_* . Full indecomposability excludes a column with only one positive entry, while a full-support column is impossible because (24) would make the sum of all row sums less than 9. Hence $1 \leq z \leq 7$.

The support has $9 - z$ rows: the distinguished row of sum $1 - a$, and $8 - z$ other rows bounded by $1/q(a)$. The remaining z row sums have total chosen to make the grand total 9. AM–GM, followed by increasing each capped row to its cap, gives

$$R \leq U_z(a) := (1 - a)q(a)^{-(8-z)} \left(\frac{8 + a - (8 - z)/q(a)}{z} \right)^z. \quad (25)$$

For completeness, the monotonicity in a capped variable x follows by differentiating the logarithm of $x[(T - x)/z]^z$, where the uncapped total has average greater than 1 while $x < 1$.

On the other hand, $\Phi(A) \geq 2 - \gamma_9$, (15), and (18) imply

$$R \geq R_0(a) := 2 - \gamma_9 + L_9(1 - a)^9 - g(h(a)). \quad (26)$$

It remains to check $R_0(a) > U_z(a)$ for $1 \leq z \leq 7$.

Here is the exact polynomial certificate. Put

$$\begin{aligned} e &= 1 - (2 - \gamma_9)a, & t &= e - \frac{(1 - \gamma_9)a}{8}, \\ Q &= (2 - \gamma_9)e^8 - t^8 = e^8 q(a), & S &= (2 - \gamma_9 + L_9(1 - a)^9)e^9 - (e + (1 - \gamma_9)a)t^8 = e^9 R_0(a), \\ W_z &= (8 + a)Q - (8 - z)e^8. \end{aligned}$$

All denominators are positive on (21), and $R_0 > U_z$ is equivalent to

$$G_z(a) := z^z Q^8 S - (1 - a)e^{9+8(8-z)} W_z^z > 0. \quad (27)$$

Each G_z has degree 82. The primary verifier expands it directly and converts it to the degree-82 Bernstein basis on $[1/500, 1/200]$; every one of the $7 \cdot 83 = 581$ coefficients is strictly positive. The smallest is approximately $3.7489581 \cdot 10^{-5}$ (the exact rational is retained by the program and logged by the audit). This contradicts (25)–(26) and excludes every marginal cut.

6 Proper tight cuts

Suppose now that $u, v \geq 1$. Define

$$\varepsilon_r = \sum_{i \in I^c} r_i - u, \quad \varepsilon_c = \sum_{j \in J^c} c_j - v.$$

For positive numbers $x_1, \dots, x_n$ of sum n and product X , binary Pinsker after grouping a subset and its complement gives

$$\left(\sum_{i \in S} x_i - |S| \right)^2 \leq -\frac{n}{2} \log X.$$

Applying this to the row and column sums, then using Cauchy–Schwarz and (2), yields

$$|\varepsilon_r| + |\varepsilon_c| \leq \sqrt{-n \log(RC)} \leq \sqrt{\frac{n\delta}{1-\delta}} =: T. \quad (28)$$

The cut identity

$$s(A[I, J]) = k - \varepsilon_r - \varepsilon_c + s(A[I^c, J^c])$$

therefore implies

$$\lambda \geq 1 - \frac{T}{k}. \quad (29)$$

For $n = 9$, the exact inequality $n\gamma_n < 1 - \gamma_n$ ensures $T < 1 \leq k$.

Because B is doubly stochastic,

$$s(B[I, J]) = k + s(B[I^c, J^c]) \geq k.$$

Tightness and (4) give the reverse bound after multiplication by λ , so equality holds and $B[I^c, J^c] = 0$. Thus B lies on the staircase face with a prescribed $u \times v$ zero block. Hwang's theorem gives

$$\text{per } B \geq \nu_{n;u,v} := \frac{\gamma_{n-u}\gamma_{n-v}}{\gamma_k}, \quad k = n - u - v, \quad (30)$$

where $\gamma_0 = 1$. The barycenter has $(n-v)!(n-u)!/k!$ supported permutations, all of the same weight; both verifiers reconstruct this count exactly.

Using (29), Bernoulli's inequality, and $1 - \delta \geq 1 - \gamma_n$ gives

$$\nu_{n;u,v} \left(1 - \frac{T}{k} \right)^n - (\gamma_n - \delta) \geq \nu_{n;u,v} - \gamma_n + \delta - \frac{n\nu_{n;u,v}}{k} \sqrt{\frac{n\delta}{1-\gamma_n}}.$$

Completing the square in $\sqrt{\delta}$ gives the uniform lower bound

$$\eta_{n;u,v} := \nu_{n;u,v} - \gamma_n - \frac{n^3 \nu_{n;u,v}^2}{4k^2(1-\gamma_n)}. \quad (31)$$

Both verifiers check $\eta_{9;u,v} > 0$ for all

$$u, v \geq 1, \quad u + v \leq 8,$$

namely all 28 proper cuts. The smallest value occurs at $(u, v, k) = (1, 7, 1)$ (and symmetrically at $(7, 1, 1)$), where

$$\eta_{9;1,7} = \frac{315}{262144}, \quad \eta_{9;1,7} = \frac{9881810825836802335}{6282434825977483864571904} > 0.$$

This contradicts $P = \gamma_n - \delta$ in every proper-cut case.

The whole, marginal, and proper cases exhaust every tight cut. No boundary global maximizer exists. Hwang's positive-support theorem identifies the sole remaining maximizer as $J_9/9$, proving Theorem 1.1.

7 Exact computational audit

No floating-point value is used in a correctness decision. The primary standard-library verifier checks:

1. the bivariate two-zero equalization identity and its sign;
2. the degree-nine permanent polynomial and the lower bound (8) by a rational Sturm sequence;
3. denominator and endpoint signs, (17), and zero Sturm roots for the degree-18 numerator on both safe intervals;
4. $q(1/500) > 1$ and all 581 exact Bernstein coefficients in (27);
5. the staircase barycenter count and all 28 values in (31).

The independent verifier imports no code from the primary verifier. It uses a literal exact Ryser sum for the two-zero matrices and Bernstein positivity on 256 subintervals. It reconstructs the marginal numerator from 19 exact values and each support polynomial from 83 exact scalar values, checks off-grid values, and proves the required positivity by independently implemented Bernstein conversion. Thus the principal expansions and both Sturm computations are not shared.

The integration test runs both programs in ordinary and optimized Python mode and requires identical output. It also inflates L_9 to a false bound and weakens the accounting of capped support rows; both independently mutated programs must reject both changes.

8 Scope and status

This note establishes dimension 9. Together with the other exact working proofs in the accompanying repository, the covered dimensions are

$$4, 5, 9, 10, 11, 12, 13, 14, 15, 16.$$

Dimensions 6, 7, 8 are not settled by these materials. The present result should be described as an exact computer-assisted working proof until the mathematical reduction and both programs have undergone independent peer review.

Reproducibility

From the n9 directory, run

```
python3 -I verify_dittert_n9.py
python3 -I audit_dittert_n9_stdlib.py
python3 -I test_n9_package.py
```

The verifier final lines must be

```
CERTIFIED
INDEPENDENT AUDIT CERTIFIED
```

and the tests must report OK.

References

- [1] G.-S. Cheon and I. M. Wanless, *Some results towards the Dittert conjecture on permanents*, Linear Algebra Appl. 436 (2012), 791–801. doi:10.1016/j.laa.2010.08.041.
- [2] G. P. Egorychev, *Solution of the van der Waerden problem for permanents*, Soviet Math. Dokl. 23 (1981), 619–622.
- [3] D. I. Falikman, *A proof of van der Waerden’s conjecture on the permanent of a stochastic matrix*, Mat. Zametki 29 (1981), 931–938.
- [4] S.-G. Hwang, *Minimum permanent on faces of staircase type of the polytope of doubly stochastic matrices*, Linear Multilinear Algebra 18 (1985), 271–306. doi:10.1080/03081088508817694.
- [5] S.-G. Hwang, *A note on a conjecture on permanents*, Linear Algebra Appl. 76 (1986), 31–44. doi:10.1016/0024-3795(86)90212-0.
- [6] C.-K. Li, *On certain convex matrix sets*, Discrete Math. 79 (1990), 323–326.
- [7] H. Minc, *Minimum permanents of doubly stochastic matrices with prescribed zero entries*, Linear Multilinear Algebra 15 (1984), 225–243. doi:10.1080/03081088408817592.
- [8] K. Pula, S.-Z. Song, and I. M. Wanless, *Minimum permanents on two faces of the polytope of doubly stochastic matrices*, Linear Algebra Appl. 434 (2011), 232–238. doi:10.1016/j.laa.2010.08.022.